TEDTRONICS · PERFBOARD FALLBACK

CP00 — First light

Win: the ESP32 switches the real four-column LED loop.

85	44	1.7 m
SOLDER POINTS	WIRES TO SOLDER	WIRE TO CUT

9	24	5
JUMPERS	AWG WIRE	GROUPS

WARNING

Disconnect USB and all power sources before you build. Do not connect the final supply or the matrix in CP00.

TOOLS

- [ ] Soldering iron and solder
- [ ] Wire stripper
- [ ] USB cable
- [ ] Wire cutter
- [ ] Multimeter
- [ ] Sleeves for wire crossings

WIRE

- Insulated 24 AWG for all hook-up wires. Run them under the board.
- Bare wire for the 20 node straps*.

* The guidelines give a size only for insulated wire. Use 24 AWG bare wire and confirm.

BUILD ORDER

1. **Pogo contact columns**
   4 parts · 20 wires · 24 solder points
2. **NODE header and wires**
   1 parts · 4 wires · 14 solder points
3. **ESP32 controller**
   3 parts · 11 wires · 47 solder points
4. **Breadboard USB switch**
   8 parts · 9 jumpers · no soldering
5. **Fit the kit and test**
   4 kit parts · 6 checks · film

Each group has three pages: views, full schematic, then parts and steps.

KEY



Red pin
Pin 1 of a part. Match it to pin 1 on the part.



Open pin
Any other pin. Top view only.



Solder pad
Bottom view. Make a solder joint here.



Insulated wire
Under the board. Solder only the two ends. Pu
crossing.



Bare strap
Joins the pins of one column. Touch no other c



Part body
Top view. Colored = this group. Dashed gray =



Row tie
Breadboard. The row already joins these holes

TWO VIEWS OF A PERFBOARD

Numbers on the edges are the **top** hole numb
same on both views.

- **TOP**: look at the part side.
- **BOTTOM (mirrored)**: turn the board over
  Column 1 is now on the right.

WORDS

WARNING

A warning tells you about a risk to people o
Read it first.

CAUTION

A caution tells you about a risk to a part or

NOTE

A note gives extra information.

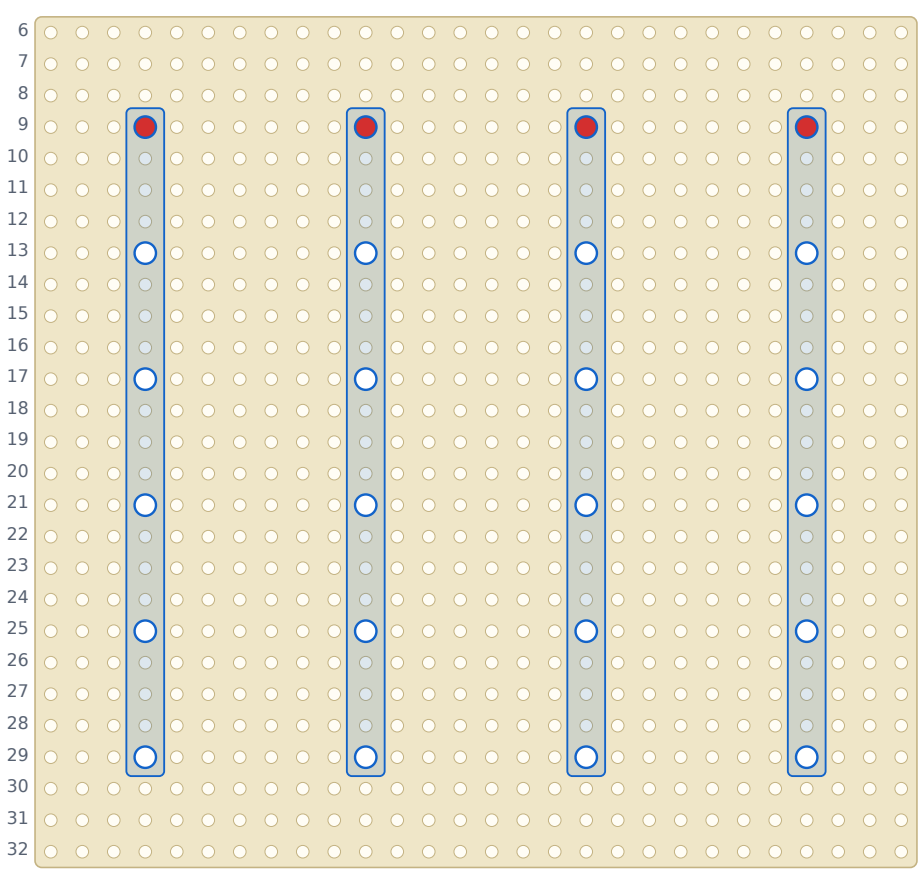
GROUP 1 OF 5 · VIEWS

Pogo contact columns

4	20	24	0.34 m
PARTS	WIRES	SOLDER POINTS	WIRE

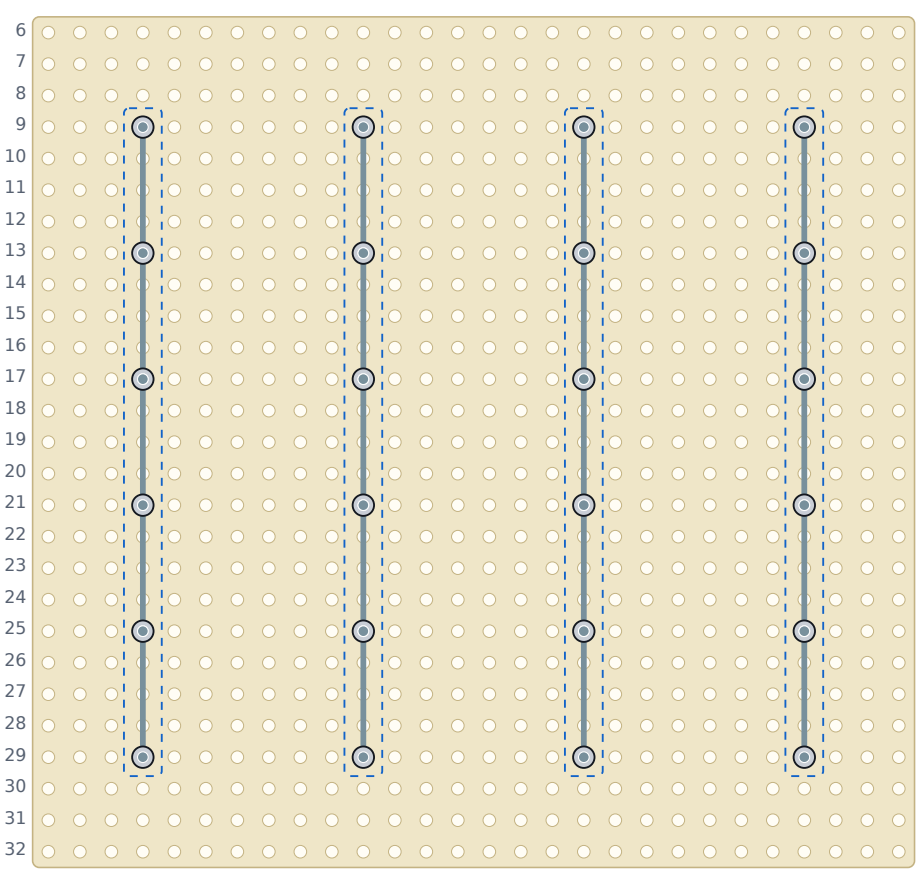
TOP

part side · red pin = pin 1



BOTTOM

mirrored · solder side · wire labels beside each wire

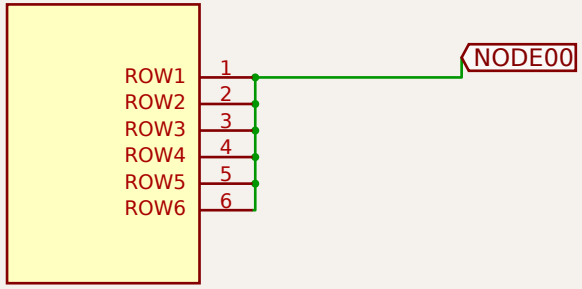


GROUP 1 OF 5 · SCHEMATIC

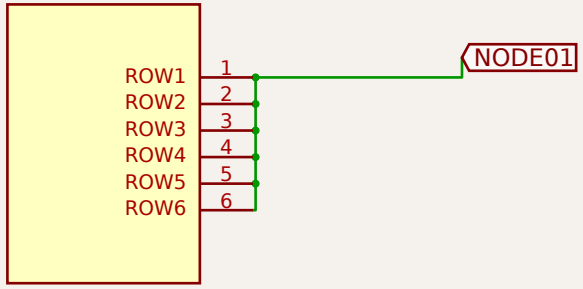
Pogo contact columns

4	20	24	0.34 m
PARTS	WIRES	SOLDER POINTS	WIRE

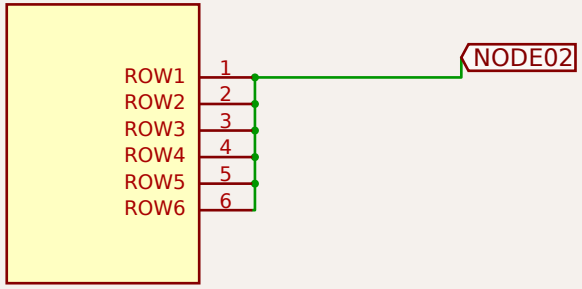
NODE00 — six contacts



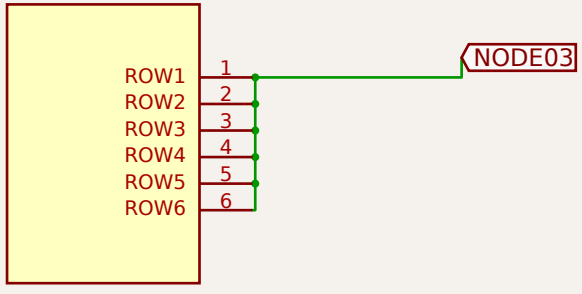
NODE01 — six contacts



NODE02 — six contacts



NODE03 — six contacts



GROUP 1 OF 5 · PARTS AND STEPS

Pogo contact columns

4	20	24	0.34 m
PARTS	WIRES	SOLDER POINTS	WIRE

COMPONENTS

REF	COMPONENT TYPE	DETAIL	PINS
J_NODE00 J_NODE01 J_NODE02 J_NODE03	**Pogo pin (spring-loaded contact)**	6 single pins, one column, 4 holes (10.16 mm) apart. The purchased pin is not identified yet. Measure it before you solder.	**4 × 6**

CUT LIST

- **20 ×** 17 mm

Bare wire, 24 AWG. 4 pad gaps (10.2 mm) plus 6 mm for the ends.

STEPS

1 **Install the pogo pins.** Put each part in from the top, pogo side up.

- [ ] J_NODE00: column 4, rows 9, 13, 17, 21, 25, 29
- [ ] J_NODE01: column 11, rows 9, 13, 17, 21, 25, 29
- [ ] J_NODE02: column 18, rows 9, 13, 17, 21, 25, 29
- [ ] J_NODE03: column 25, rows 9, 13, 17, 21, 25, 29

2 **Solder the pins.** Turn the board over. Solder all 24 pins from the bottom.

- [ ] Each pin is straight and square to the board.
- [ ] No solder bridge to the next column.

3 **Cut 20 bare straps**, each 17 mm.

4 **Solder the straps.** In each column, join pin 1–2, 2–3, 3–4, 4–5, 5–6 on the bottom.

- [ ] Column 4: 5 straps
- [ ] Column 11: 5 straps
- [ ] Column 18: 5 straps
- [ ] Column 25: 5 straps

5 **Measure with the multimeter in continuity mode.** Power OFF.

- [ ] All 6 pins in one column read below 1 Ω.
- [ ] Pins in two different columns read open (above 10 MΩ).

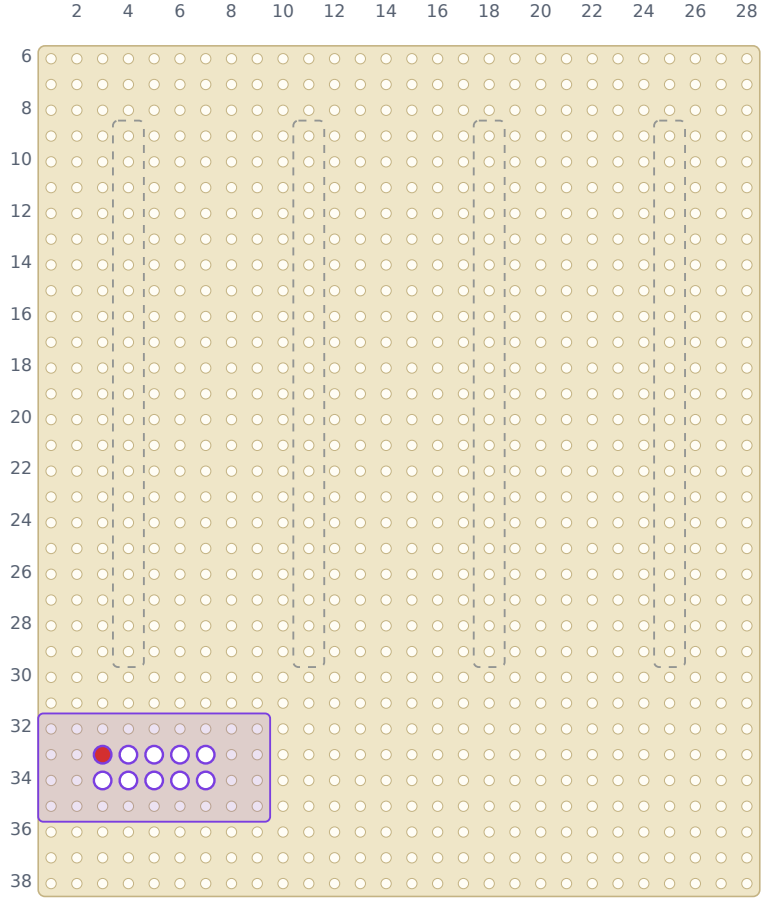
GROUP 2 OF 5 · VIEWS

NODE header and wires

1	4	14	0.41 m
PARTS	WIRES	SOLDER POINTS	WIRE

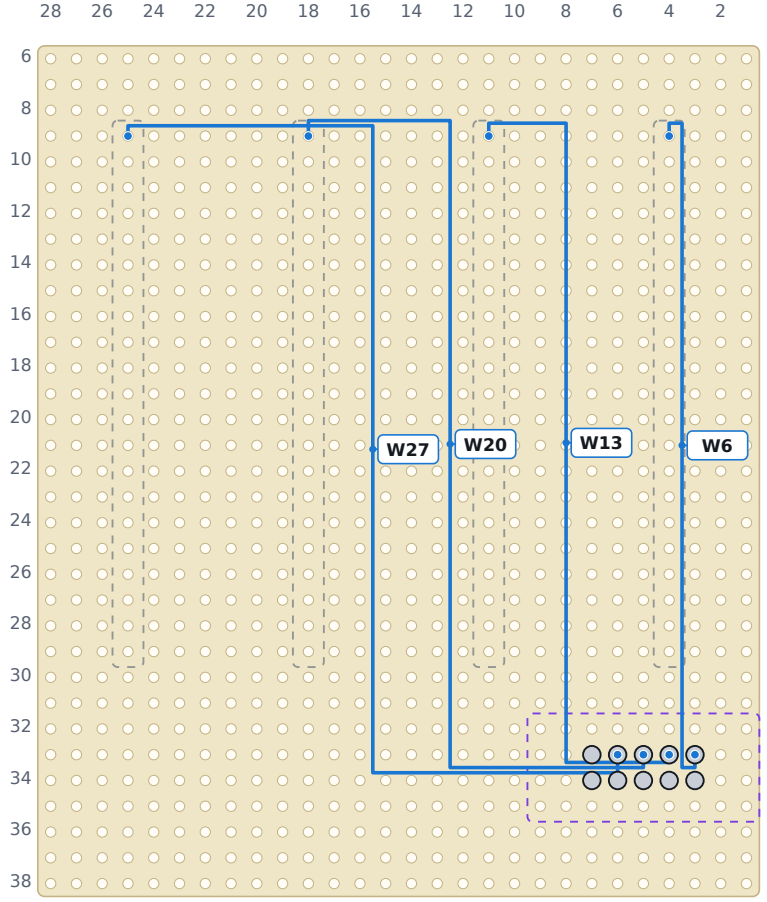
TOP

part side · red pin = pin 1



BOTTOM

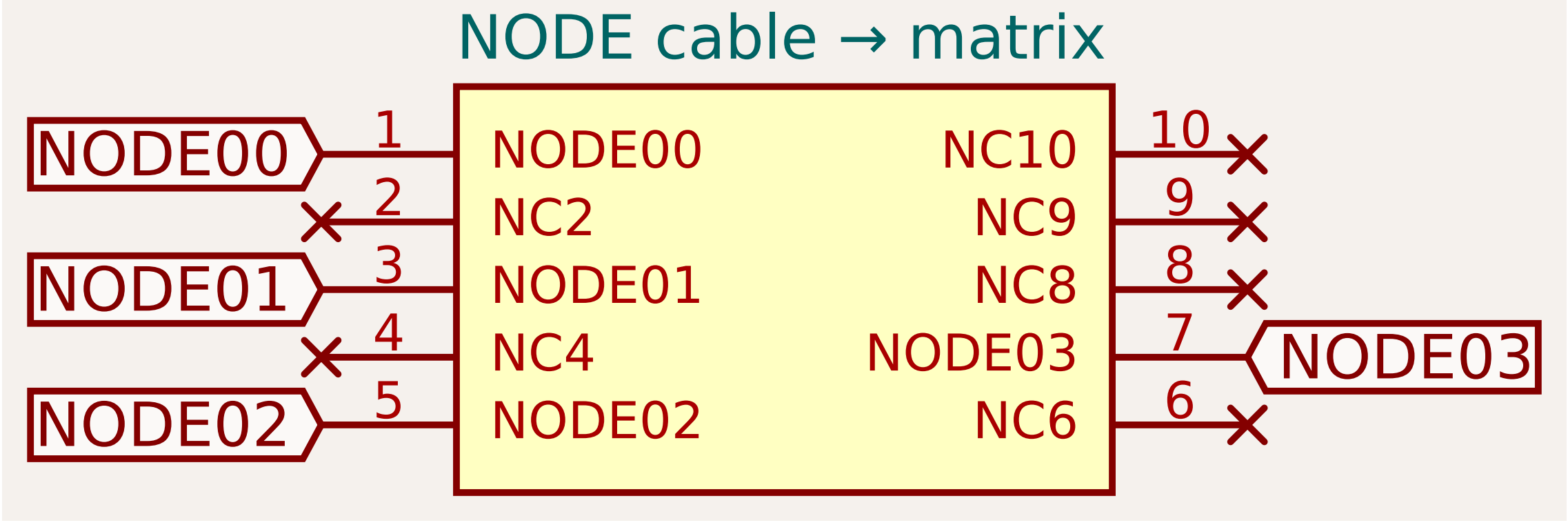
mirrored · solder side · wire labels beside each wire



GROUP 2 OF 5 · SCHEMATIC

NODE header and wires

1	4	14	0.41 m
PARTS	WIRES	SOLDER POINTS	WIRE



GROUP 2 OF 5 · PARTS AND STEPS

NODE header and wires

1	4	14	0.41 m
PARTS	WIRES	SOLDER POINTS	WIRE

COMPONENTS

REF	COMPONENT TYPE	DETAIL	PINS
J_NODE	**IDC box header, 2 × 5, 2.54 mm**	Bottom side. Odd pins on row 33, even pins on row 34. Only pins 1, 3, 5, 7 carry signals; pins 2, 4, 6, 8, 9, 10 stay unconnected. Guidelines: drill keyed IDC positions to 1.2 mm first.	**10**

J_NODE pin map

1	NODE00	col 3, row 33
2	NC2	col 3, row 34
3	NODE01	col 4, row 33
4	NC4	col 4, row 34
5	NODE02	col 5, row 33
6	NC6	col 5, row 34
7	NODE03	col 6, row 33
8	NC8	col 6, row 34
9	NC9	col 7, row 33
10	NC10	col 7, row 34

CUT LIST

- **1** × 79 mm
- **1** × 94 mm
- **1** × 110 mm
- **1** × 125 mm

Insulated 24 AWG. Routed length plus 10 mm for the ends.

WIRES

ID		FROM → TO	CUT
W6		J_NODE00.1 → J_NODE.1 · (4,9) → (3,33)	**79 mm**
W13		J_NODE01.1 → J_NODE.3 · (11,9) → (4,33)	**94 mm**
W20		J_NODE02.1 → J_NODE.5 · (18,9) → (5,33)	**110 mm**
W27		J_NODE03.1 → J_NODE.7 · (25,9) → (6,33)	**125 mm**

STEPS

1 **Install the J_NODE header** on the bottom side. Pin 1 at column 3, row 33.

- [ ] Red pad is at pin 1.

2 **Solder the 10 header pins** from the top side.

- [ ] No solder bridge between pins.

3 **Cut 4 wires:** **1** × 79 mm · **1** × 94 mm · **1** × 110 mm · **1** × 125 mm

4 **Route and solder the wires on the bottom.** Solder only the two ends of each wire.

- [ ] `W6`: `J_NODE00.1` → `J_NODE.1`, 🔵 79 mm
- [ ] `W13`: `J_NODE01.1` → `J_NODE.3`, 🔵 94 mm
- [ ] `W20`: `J_NODE02.1` → `J_NODE.5`, 🔵 110 mm
- [ ] `W27`: `J_NODE03.1` → `J_NODE.7`, 🔵 125 mm

5 **Put a sleeve on every crossing.** A crossing is not a connection.

6 **Measure.** Power OFF.

- [ ] Each wire reads below 1 Ω from end to end.
- [ ] Two different wires read open.

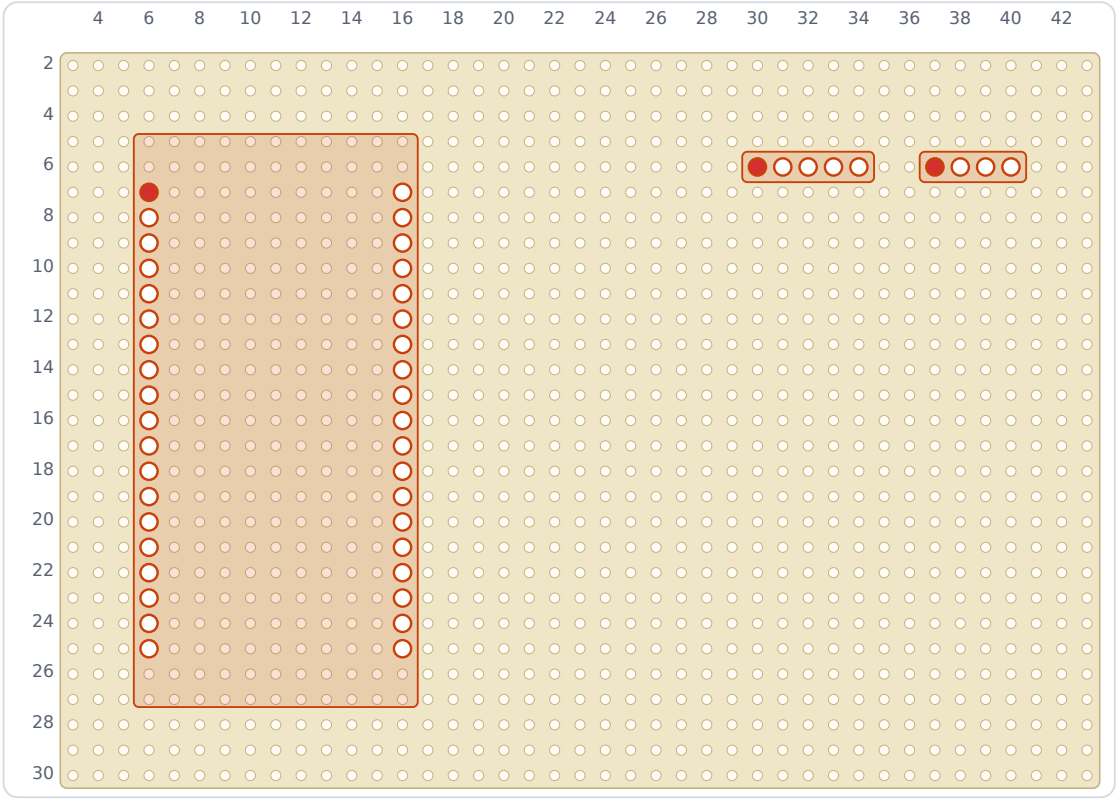
GROUP 3 OF 5 · VIEWS

ESP32 controller

3	11	47	0.94 m
PARTS	WIRES	SOLDER POINTS	WIRE

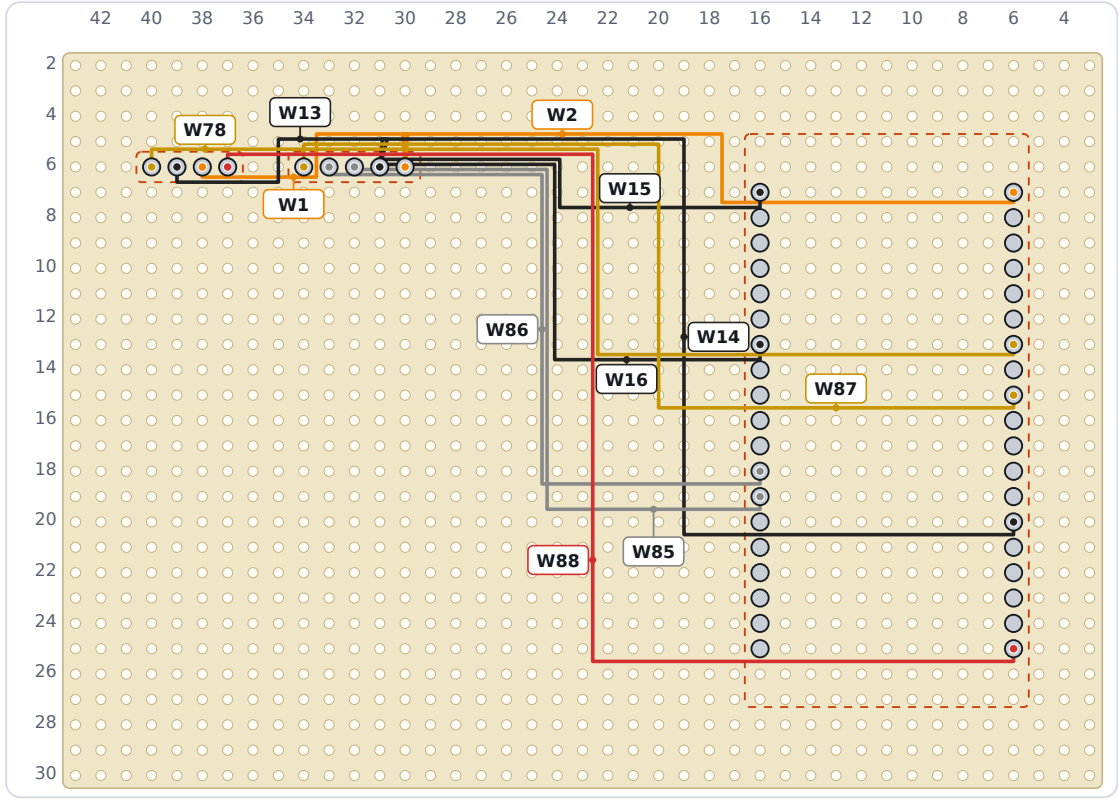
TOP

part side · red pin = pin 1



BOTTOM

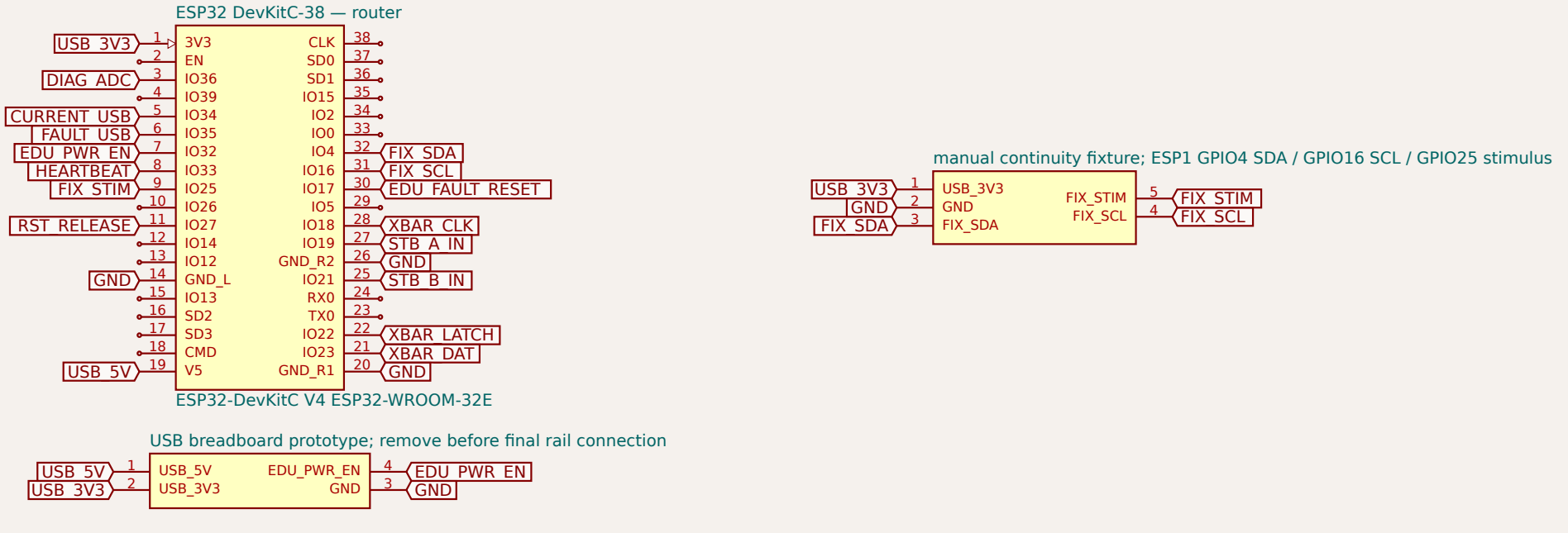
mirrored · solder side · wire labels beside each wire



GROUP 3 OF 5 · SCHEMATIC

ESP32 controller

3	11	47	0.94 m
PARTS	WIRES	SOLDER POINTS	WIRE



GROUP 3 OF 5 · PARTS AND STEPS

ESP32 controller

3	11	47	0.94 m
PARTS	WIRES	SOLDER POINTS	WIRE

COMPONENTS

REF	COMPONENT TYPE	DETAIL	PINS
U_ESP1	**ESP32 DevKitC V4 board, 2 × 19 pins**	Two pin rows, 10 holes (25.4 mm) apart. Module ESP32-WROOM-32E. Check your pinout against the labels before you solder. ESP32-DevKitC V4 ESP32-WROOM-32E	**38**
J_FIXTURE	**Straight pin header, 1 × 5, 2.54 mm**	Single row, top side. Pin 1 at the left.	**5**
J_PROTOTYPE	**Straight pin header, 1 × 4, 2.54 mm**	Single row, top side. Pin 1 at the left.	**4**

U_ESP1 pin map · wired pins only (9 of 38)

1	3V3	col 6, row 7
7	IO32	col 6, row 13
9	IO25	col 6, row 15
14	GND_L	col 6, row 20
19	V5	col 6, row 25
20	GND_R1	col 16, row 7
26	GND_R2	col 16, row 13
31	IO16	col 16, row 18
32	IO4	col 16, row 19

J_FIXTURE pin map

1	USB_3V3	col 30, row 6
2	GND	col 31, row 6
3	FIX_SDA	col 32, row 6
4	FIX_SCL	col 33, row 6
5	FIX_STIM	col 34, row 6

J_PROTOTYPE pin map

1	USB_5V	col 37, row 6
2	USB_3V3	col 38, row 6
3	GND	col 39, row 6
4	EDU_PWR_EN	col 40, row 6

CUT LIST

- **1 × 36 mm**
- **1 × 36 mm**
- **1 × 56 mm**
- **1 × 71 mm**
- **1 × 79 mm**
- **2 × 89 mm**
- **1 × 110 mm**
- **1 × 115 mm**
- **1 × 120 mm**
- **1 × 143 mm**

Insulated 24 AWG. Routed length plus 10 mm for the ends.

WIRES

ID		FROM → TO	CUT
W1		J_FIXTURE.1 → J_PROTOTYPE.2 · (30,6) → (38,6)	**36 mm**
W2		J_FIXTURE.1 → U_ESP1.1 · (30,6) → (6,7)	**79 mm**
W13		J_FIXTURE.2 → J_PROTOTYPE.3 · (31,6) → (39,6)	**36 mm**

ID	FROM → TO	CUT
W14 ●	J_FIXTURE.2 → U_ESP1.14 · (31,6) → (6,20)	**115 mm**
W15 ●	J_FIXTURE.2 → U_ESP1.20 · (31,6) → (16,7)	**56 mm**
W16 ●	J_FIXTURE.2 → U_ESP1.26 · (31,6) → (16,13)	**71 mm**
W78 ●	J_PROTOTYPE.4 → U_ESP1.7 · (40,6) → (6,13)	**120 mm**
W85 ●	J_FIXTURE.3 → U_ESP1.32 · (32,6) → (16,19)	**89 mm**
W86 ●	J_FIXTURE.4 → U_ESP1.31 · (33,6) → (16,18)	**89 mm**
W87 ●	J_FIXTURE.5 → U_ESP1.9 · (34,6) → (6,15)	**110 mm**
W88 ●	J_PROTOTYPE.1 → U_ESP1.19 · (37,6) → (6,25)	**143 mm**

STEPS

1 **Install the ESP32 DevKit.**

- [ ] U_ESP1: column 6, rows 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25; column 16, rows 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25

2 **Install the two headers** on the top side. Red pad is at pin 1.

- [ ] J_FIXTURE: column 30, rows 6; column 31, rows 6; column 32, rows 6; column 33, rows 6; column 34, rows 6
- [ ] J_PROTOTYPE: column 37, rows 6; column 38, rows 6; column 39, rows 6; column 40, rows 6

3 **Solder all 47 pins** from the bottom.

- [ ] No solder bridge between pins.

4 **Cut 11 wires** from the cut list.

5 **Route and solder the wires on the bottom.** Solder only the two ends. Put a sleeve on every crossing.

- [ ] W1 ● J_FIXTURE.1 → J_PROTOTYPE.2
- [ ] W2 ● J_FIXTURE.1 → U_ESP1.1
- [ ] W13 ● J_FIXTURE.2 → J_PROTOTYPE.3
- [ ] W14 ● J_FIXTURE.2 → U_ESP1.14
- [ ] W15 ● J_FIXTURE.2 → U_ESP1.20
- [ ] W16 ● J_FIXTURE.2 → U_ESP1.26
- [ ] W78 ● J_PROTOTYPE.4 → U_ESP1.7
- [ ] W85 ● J_FIXTURE.3 → U_ESP1.32
- [ ] W86 ● J_FIXTURE.4 → U_ESP1.31
- [ ] W87 ● J_FIXTURE.5 → U_ESP1.9
- [ ] W88 ● J_PROTOTYPE.1 → U_ESP1.19

6 **Measure.** Power OFF.

- [ ] Each wire reads below 1 Ω from end to end.
- [ ] GND and 3V3 read open.

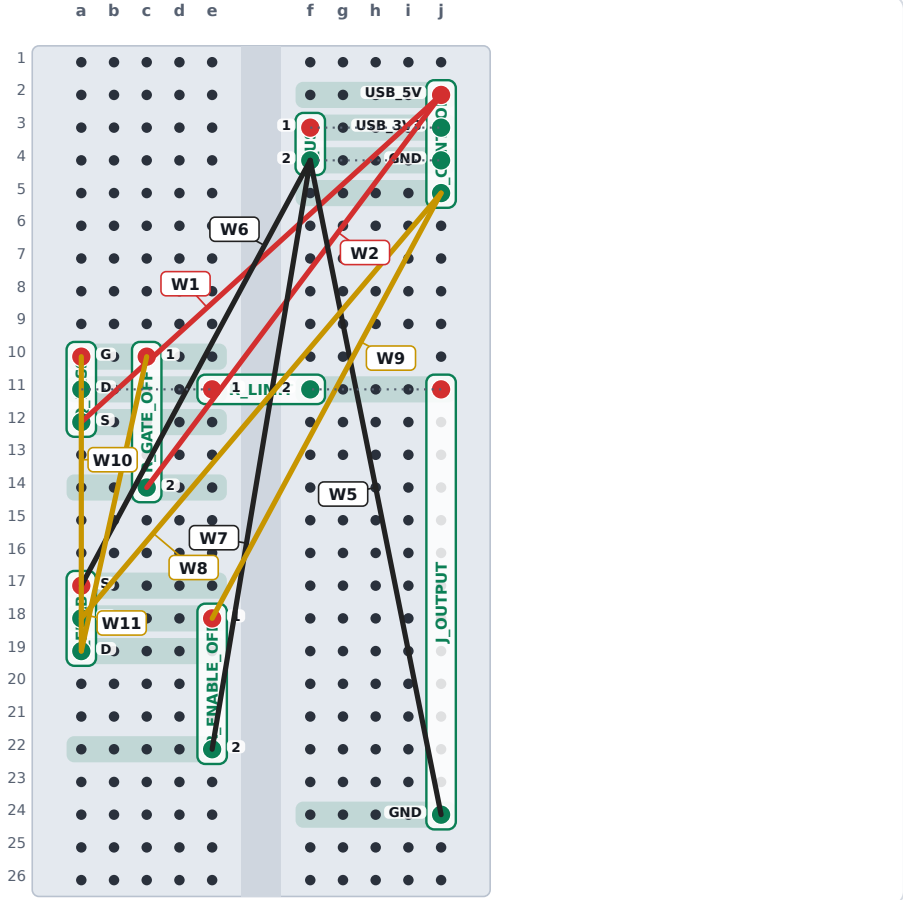
GROUP 4 OF 5 · VIEWS

Breadboard USB switch

8	9	0
PARTS	JUMPERS	SOLDER POINTS

BREADBOARD

rows a–e and f–j are joined



JUMPERS

ID	FROM → TO	CUT
W1	J_CONTROL.1 → Q_PASS.3 · j2 → a12	
W2	J_CONTROL.1 → R_GATE_OFF.2 · j2 → c14	
W5	C_USB.2 → J_OUTPUT.2 · f4 → j24	
W6	C_USB.2 → Q_ENABLE.1 · f4 → a17	
W7	C_USB.2 → R_ENABLE_OFF.2 · f4 → e22	
W8	J_CONTROL.4 → Q_ENABLE.2 · j5 → a18	
W9	J_CONTROL.4 → R_ENABLE_OFF.1 · j5 → e18	
W10	Q_ENABLE.3 → Q_PASS.1 · a19 → a10	
W11	Q_ENABLE.3 → R_GATE_OFF.1 · a19 → c10	

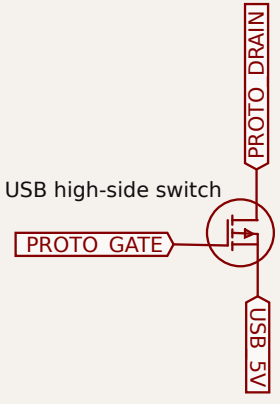
No jumper for 4 row ties. The breadboard rows already join those holes.

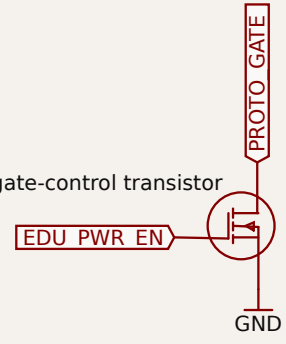
> **CAUTION**
>
> Q_PASS and Q_ENABLE have three pins in a fixed order. Read the pin letters on the drawing.

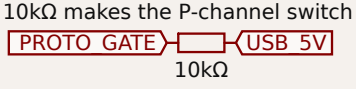
GROUP 4 OF 5 · SCHEMATIC

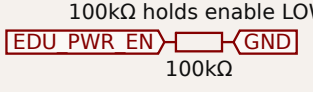
Breadboard USB switch

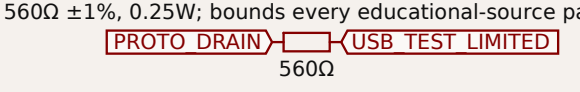
8	9	0
PARTS	JUMPERS	SOLDER POINTS

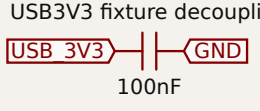


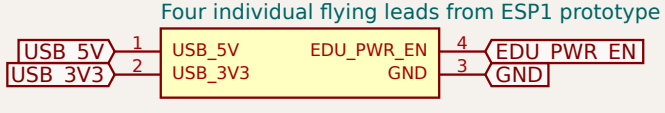


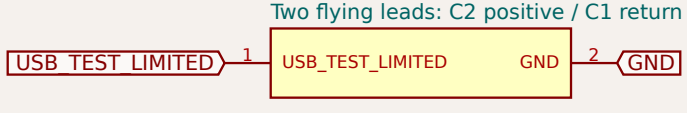












GROUP 4 OF 5 · PARTS AND STEPS

Breadboard USB switch

8	9	0
PARTS	JUMPERS	SOLDER POINTS

COMPONENTS

REF	COMPONENT TYPE	DETAIL	PINS
J_CONTROL	**Male flying jumper leads, 4 pcs, 24 AWG**	Individual leads, not a rigid plug. Purchased set not identified yet.	4
Q_PASS	**P-channel MOSFET**	Pins: gate, drain, source. NDP6020P	3
Q_ENABLE	**N-channel MOSFET**	Pins: source, gate, drain. 2N7000	3
R_GATE_OFF	**Axial resistor, 1/4 W, 1 %** 10k	MFR-25FBF52-10K	2
R_ENABLE_OFF	**Axial resistor, 1/4 W, 1 %** 100k	MFR-25FBF52-100K	2
R_LIMIT	**Axial resistor, 1/4 W, 1 %** 560	Crosses the breadboard center channel. MFR-25FBF52-560R	2
J_OUTPUT	**Male flying jumper leads, 2 pcs, 24 AWG**	Individual leads, not a rigid plug. Purchased set not identified yet.	2
C_USB	**Ceramic capacitor** 100nF	C315C104K5R5TA	2

J_CONTROL pin map

1	USB_5V	j2
2	USB_3V3	j3
3	GND	j4
4	EDU_PWR_EN	j5

J_OUTPUT pin map

1	USB_TEST_LIMITED	j11
2	GND	j24

WIRES

ID		FROM → TO	CUT
W1	red	J_CONTROL.1 → Q_PASS.3 · j2 → a12	
W2	red	J_CONTROL.1 → R_GATE_OFF.2 · j2 → c14	
W5	black	C_USB.2 → J_OUTPUT.2 · f4 → j24	
W6	black	C_USB.2 → Q_ENABLE.1 · f4 → a17	
W7	black	C_USB.2 → R_ENABLE_OFF.2 · f4 → e22	
W8	yellow	J_CONTROL.4 → Q_ENABLE.2 · j5 → a18	
W9	yellow	J_CONTROL.4 → R_ENABLE_OFF.1 · j5 → e18	
W10	yellow	Q_ENABLE.3 → Q_PASS.1 · a19 → a10	
W11	yellow	Q_ENABLE.3 → R_GATE_OFF.1 · a19 → c10	

STEPS

1 **Put the parts in the holes.** Pin letters on the drawing must match.

- [ ] J_CONTROL: j2, j3, j4, j5
- [ ] Q_PASS: a10, a11, a12
- [ ] Q_ENABLE: a17, a18, a19
- [ ] R_GATE_OFF: c10, c14
- [ ] R_ENABLE_OFF: e18, e22
- [ ] R_LIMIT: e11, f11
- [ ] J_OUTPUT: j11, j24
- [ ] C_USB: f3, f4

2 **Add the 9 jumpers.** Use male jumper wire. No soldering.

- [ ] W1 🔴 j2 → a12
- [ ] W2 🔴 j2 → c14
- [ ] W5 ⚫ f4 → j24
- [ ] W6 ⚫ f4 → a17
- [ ] W7 ⚫ f4 → e22
- [ ] W8 🟡 j5 → a18
- [ ] W9 🟡 j5 → e18
- [ ] W10 🟡 a19 → a10
- [ ] W11 🟡 a19 → c10

3 **Check the 4 row ties.** USB disconnected.

- [ ] Each tie reads below 1 Ω.

4 **Check pin 1 and polarity** of every part.

GROUP 5 OF 5

Fit the kit and test

4	6	1
KIT PARTS	CHECKS	FILM

KIT PARTS ON BOARD A1

ROW	PART	ON
1 · y 9	Fake battery	− C1, + C2
2 · y 13	Resistor 1 kΩ	C2 – C3
3 · y 17	LED	anode C3, cathode C4
4 · y 21	Kit jumper	C4 → C1

C1–C4 are columns 1–4 (NODE00–NODE03).

STEPS

1. **Link the test source to A1.** Use two flying leads from the breadboard J_OUTPUT.

   - ☐ J_OUTPUT pin 1 → A1 J_NODE pin 3 (C2, positive).
   - ☐ J_OUTPUT pin 2 → A1 J_NODE pin 1 (C1, ground).

2. **Connect the USB prototype leads.** Four separate leads from matrix J_PROTOTYPE to breadboard J_CONTROL.

   - ☐ Pin 1 matches pin 1 on both ends.

3. **Fit the four kit parts** on the contact rows. See the table above.

   - ☐ LED polarity is correct.
   - ☐ Measured resistor value is 1 kΩ.

CHECKS

1. **Identify the hardware.** Find the ESP32 DevKit pinout, USB 5V and 3V3. Measure the real pogo and adapter size before final solder.

   - ☐ Pinout matches the table.

2. **Measure A1, unpowered.** Pins in one column: below 1 Ω. Different columns: above 10 MΩ.

   Column result _____ pass / fail

3. **Measure the prototype, A1 disconnected.** USB 5V: 5.25 V or less. 3V3: 3.20 to 3.38 V. Short-circuit current: 9.5 mA or less.

   USB 5V _____ V
   3V3 _____ V
   Short current _____ mA

CAUTION

Short the output only for the current test. Remove the short before you fit the kit.

4 **Run the firmware.** GPIO32 LOW: LED is OFF. Enable: LED is ON. Disable: LED is OFF.

Firmware revision _____

Voltage at C2, under load _____ V

NOTE

The test source is not regulated. Do not call it the final 5V rail.

5 **Do the negative test.** Reset the ESP32. The LED must go dark. Do not touch the kit.

- [ ] LED is dark after reset.

6 **Film the win.** Show power OFF and the firmware revision. Show the LED switch ON and OFF. Show the real readings.

- [ ] Record readings, units, equipment, link positions and firmware revision.

- [ ] **Accept CP00 — all checks pass**